
Lab Assignment # 17

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Task 1 – Password Security Enhancement

Analyse an AI-generated Python program that stores passwords in plain text and improve its security.

Instructions:

- Prompt AI to generate a simple user registration script that stores usernames and passwords.
- Check whether passwords are stored as plain text.
- Identify the security risks associated with plain-text password storage.
- Refactor the program to use secure password hashing (e.g., hashlib or bcrypt).
- Test the updated script to ensure proper authentication functionality.

Expected Output:

A secure Python script that stores hashed passwords instead of plain text and successfully validates user login credentials.

Prompt:

Generate a python code that cleans and prepare a student academic dataset for analysis. The dataset contains follwing columns: student_id, name, gender, department, marks, attendance, gender_encoded, department_encoded. The code should handle missing values in marks, attendance, and department. remove duplicates, standardize text fields such as department names, and categorical variable like gender and department. Finally, the code should output a cleaned and encoded pandas Dataframe ready for analysis.

Code:

```
# Task-01
# Generate a python code that cleans and prepare a students academic dataset
import pandas as pd
from sklearn.preprocessing import LabelEncoder

# Load dataset
data = pd.read_csv(r"C:\AIAC LAB\students_cleaned.csv")
print("Original data shape:", data.shape)
print(data.head())

# Handle missing values
data['marks'].fillna(data['marks'].mean(), inplace=True)
data['attendance'].fillna(data['attendance'].mean(), inplace=True)
data['department'].fillna(data['department'].mode()[0], inplace=True)

# Remove duplicates
data.drop_duplicates(inplace=True)

# Standardize text fields
data['department'] = data['department'].str.strip().str.title()
data['gender'] = data['gender'].str.strip().str.lower()

# Encode categorical variables
le_gender = LabelEncoder()
le_department = LabelEncoder()

data['gender_encoded'] = le_gender.fit_transform(data['gender'])
data['department_encoded'] = le_department.fit_transform(data['department'])

# Display cleaned data
print("\nCleaned data shape:", data.shape)
print(data.head())
print("\nData types:\n", data.dtypes)
print("\nMissing values:\n", data.isnull().sum())
```

Output:

```
Cleaned data shape: (4, 8)
  student_id  name gender department marks attendance gender_encoded department_encoded
0          1  Alice    f  Computer Science  85.0      95.0           0              0
1          2   Bob    m    Electronics  88.5      85.0           1              1
2          3  Charlie NaN  Computer Science  92.0      90.0           2              0
3          4   David    m  Mechanical Engineering  88.5      90.0           1              2

Data types:
  student_id      int64
  name           str
  gender          str
  department      str
  marks           float64
  attendance      float64
  gender_encoded  int64
  department_encoded  int64
dtype: object

Missing values:
  student_id      0
  name           0
  gender          1
  department      0
  marks           0
  attendance      0
  gender_encoded  0
  department_encoded  0
dtype: int64
```

Justification:

The code performs several essential data preprocessing steps to ensure the dataset is clean and ready for analysis. It handles missing values by filling them with appropriate statistics (mean for numerical columns and mode for categorical columns), removes duplicates to avoid bias in analysis, standardizes text fields to maintain consistency, and encodes categorical variables to make them suitable for machine learning algorithms. Finally, it provides a summary of the cleaned dataset, including its shape, data types, and any remaining missing values.

Task 2 – Secure API Key Management

Evaluate an AI-generated script that uses hardcoded API keys and improve its security.

Instructions:

- Ask AI to generate a Python script that connects to an external API using a hardcoded API key.
- Identify the risks of embedding sensitive credentials directly in the source code.
- Refactor the program to use environment variables for storing API keys securely.
- Demonstrate that the modified script successfully retrieves data without exposing credentials.

Expected Output:

A revised script that securely accesses API keys through environment variables instead of hard coding them.

Prompt:

Generate a python code that performs preprocess financial transaction data. convert transaction data columns to datetime format, derive new features such as transaction month, day of the week, and year. Normalize transaction amounts column and handle extreme transaction values by capping them at the 95th percentile. Finally, the code should output a preprocessed pandas DataFrame ready for analysis.

Code:

```
# Task-02
# Generate a python code that performs preprocess financial transaction data. convert tran
import pandas as pd
from sklearn.preprocessing import MinMaxScaler

# Load dataset
transactions = pd.read_csv(r"C:\AIAC LAB\transactions_raw.csv")
print("Original data shape:", transactions.shape)
print(transactions.head())

# Convert to datetime format
transactions['transaction_date'] = pd.to_datetime(transactions['transaction_date'])

# Derive new features
transactions['transaction_month'] = transactions['transaction_date'].dt.month
transactions['transaction_day_of_week'] = transactions['transaction_date'].dt.day_name()
transactions['transaction_year'] = transactions['transaction_date'].dt.year

# Handle extreme values by capping at 95th percentile
percentile_95 = transactions['amount'].quantile(0.95)
transactions['amount'] = transactions['amount'].clip(upper=percentile_95)

# Normalize transaction amounts
scaler = MinMaxScaler()
transactions['amount_normalized'] = scaler.fit_transform(transactions[['amount']])

# Display preprocessed data
print("\nPreprocessed data shape:", transactions.shape)
print(transactions.head())
print("\nData types:\n", transactions.dtypes)
print("\nMissing values:\n", transactions.isnull().sum())
```

Output:

```
Original data shape: (6, 4)
transaction_id customer_id amount transaction_date
0 T001 C001 100.5 2025-01-15
1 T002 C002 250.0 2025-02-20
2 T003 C001 50.0 2025-01-10
3 T004 C003 10000.0 2025-03-05
4 T005 C002 175.5 2025-02-25

Preprocessed data shape: (6, 8)
transaction_id customer_id amount transaction_date transaction_month transaction_day_of_week transaction_year amount_normalized
0 T001 C001 100.5 2025-01-15 1 Wednesday 2025 0.006722
1 T002 C002 250.0 2025-02-20 2 Thursday 2025 0.026622
2 T003 C001 50.0 2025-01-10 1 Friday 2025 0.000000
3 T004 C003 7562.5 2025-03-05 3 Wednesday 2025 1.000000
4 T005 C002 175.5 2025-02-25 2 Tuesday 2025 0.016705

Data types:
transaction_id str
customer_id str
amount float64
transaction_date datetime64[us]
transaction_month int32
transaction_day_of_week str
transaction_year int32
amount_normalized float64
dtype: object

Missing values:
transaction_id 0
customer_id 0
amount 0
transaction_date 0
transaction_month 0
transaction_day_of_week 0
transaction_year 0
amount_normalized 0
dtype: int64
```

Justification:

The code performs the required preprocessing steps, including handling missing values, removing duplicates, standardizing text fields, encoding categorical variables, converting date formats, deriving new features, capping extreme values, and normalizing the transaction amounts. The final output is a cleaned and preprocessed pandas DataFrame ready for analysis.

Task 3 – Weak Encryption Detection

Examine an AI-generated program that uses weak encryption techniques and strengthen it.

Instructions:

- Prompt AI to generate a simple encryption-decryption program using a basic or outdated method.
- Identify weaknesses in the encryption approach (e.g., reversible encoding or outdated algorithms).
- Refactor the program using stronger encryption techniques available in Python libraries.
- Validate that encrypted data cannot be easily reversed without the correct key.

Expected Output:

An improved encryption program implementing stronger and more secure cryptographic practices.

Prompt:

Generate a python code that clean and preprocess environmental sensor data. handle missing values in temperature, humidity, and air quality index columns by using interpolation. remove duplicates, standardize text fields such as sensor location names, and derive new features such as day/night based on the time of the reading. Finally, the code should output a cleaned and preprocessed pandas DataFrame ready for analysis.

Code:

```
# Task-03
# Generate a python code that clean and preprocess environmental sensor data. handle
import pandas as pd
import numpy as np

# Load dataset
data = pd.read_csv(r"C:\AIAC LAB\sensor_data_raw_detailed.csv")
print("Original data shape:", data.shape)
print(data.head())

# Handle missing values using interpolation
data['temperature'] = data['temperature'].interpolate(method='linear')
data['humidity'] = data['humidity'].interpolate(method='linear')
data['air_quality_index'] = data['air_quality_index'].interpolate(method='linear')

# Remove duplicates
data.drop_duplicates(inplace=True)

# Standardize text fields
data['sensor_location'] = data['sensor_location'].str.strip().str.title()

# Convert timestamp to datetime
data['timestamp'] = pd.to_datetime(data['timestamp'])

# Derive day/night feature based on hour
data['hour'] = data['timestamp'].dt.hour
data['day_night'] = data['hour'].apply(lambda x: 'Night' if 6 <= x < 18 else 'Day')

# Display cleaned data
print("\nCleaned data shape:", data.shape)
print(data.head())
print("\nData types:\n", data.dtypes)
print("\nMissing values:\n", data.isnull().sum())
```

Output:

```

traffic = pd.read_csv(r"C:\AIAC LAB\traffic_data_raw.csv")
Original data shape: (208, 6)

```

	timestamp	sensor_id	sensor_location	temperature	humidity	air_quality_index
0	2025-01-01 00:00:00	S3	NaN	24.627189	81.271645	37.489499
1	2025-01-01 00:30:00	S4	Residential	16.256397	NaN	59.092204
2	2025-01-01 01:00:00	S1	NaN	17.266938	57.209033	42.947044
3	2025-01-01 01:30:00	S3	NaN	23.979983	54.817752	81.036983
4	2025-01-01 02:00:00	S3	Industrial Area	NaN	53.765060	42.451622

```

Cleaned data shape: (205, 8)

```

	timestamp	sensor_id	sensor_location	temperature	humidity	air_quality_index	hour	day_night
0	2025-01-01 00:00:00	S3	NaN	24.627189	81.271645	37.489499	0	Day
1	2025-01-01 00:30:00	S4	Residential	16.256397	69.240339	59.092204	0	Day
2	2025-01-01 01:00:00	S1	NaN	17.266938	57.209033	42.947044	1	Day
3	2025-01-01 01:30:00	S3	NaN	23.979983	54.817752	81.036983	1	Day
4	2025-01-01 02:00:00	S3	Industrial Area	21.615641	53.765060	42.451622	2	Day

```

Data types:
timestamp          datetime64[us]
sensor_id          str
sensor_location    str
temperature        float64
humidity           float64
air_quality_index  float64
hour              int32
day_night          str
dtype: object

Missing values:
timestamp          0
sensor_id          0
sensor_location    31
temperature        0
humidity           0

```

2	2025-01-01 01:00:00	S1	NaN	17.266938	57.209033	42.947044	1	Day
3	2025-01-01 01:30:00	S3	NaN	23.979983	54.817752	81.036983	1	Day
4	2025-01-01 02:00:00	S3	Industrial Area	21.615641	53.765060	42.451622	2	Day

```

Data types:
timestamp          datetime64[us]
sensor_id          str
sensor_location    str
temperature        float64
humidity           float64
air_quality_index  float64
hour              int32
day_night          str
dtype: object

Missing values:
timestamp          0
sensor_id          0
sensor_location    31
temperature        0
humidity           0

```

	timestamp	sensor_id	sensor_location	temperature	humidity	air_quality_index	hour	day_night
0	2025-01-01 00:00:00	S3	NaN	24.627189	81.271645	37.489499	0	Day
1	2025-01-01 00:30:00	S4	Residential	16.256397	69.240339	59.092204	0	Day
2	2025-01-01 01:00:00	S1	NaN	17.266938	57.209033	42.947044	1	Day
3	2025-01-01 01:30:00	S3	NaN	23.979983	54.817752	81.036983	1	Day
4	2025-01-01 02:00:00	S3	Industrial Area	21.615641	53.765060	42.451622	2	Day

```

Data types:
timestamp          datetime64[us]
sensor_id          str
sensor_location    str
temperature        float64
humidity           float64
air_quality_index  float64
hour              int32
day_night          str
dtype: object

Missing values:
timestamp          0
sensor_id          0
sensor_location    31
temperature        0
humidity           0

```

	timestamp	sensor_id	sensor_location	temperature	humidity	air_quality_index	hour	day_night
0	2025-01-01 00:00:00	S3	NaN	24.627189	81.271645	37.489499	0	Day
1	2025-01-01 00:30:00	S4	Residential	16.256397	69.240339	59.092204	0	Day
2	2025-01-01 01:00:00	S1	NaN	17.266938	57.209033	42.947044	1	Day
3	2025-01-01 01:30:00	S3	NaN	23.979983	54.817752	81.036983	1	Day
4	2025-01-01 02:00:00	S3	Industrial Area	21.615641	53.765060	42.451622	2	Day

```

Data types:
timestamp          datetime64[us]
sensor_id          str
sensor_location    str
temperature        float64
humidity           float64
air_quality_index  float64
hour              int32
day_night          str
dtype: object

Missing values:
timestamp          0
sensor_id          0
sensor_location    31
temperature        0
humidity           0

```

Justification:

Interpolation is a suitable method for handling missing values in time-series sensor data as it estimates missing values based on existing data points, preserving the temporal structure of the data. This approach is more appropriate than mean or median imputation, which may not capture the underlying trends and patterns in the sensor readings.

Task 4 – Real-Time Application: Security Review of AI-Generated Web Service Scenario:

Students select an AI-generated web service snippet (e.g., REST API, file upload handler, or authentication module).

Instructions:

- Perform a structured security review of the generated code.
- Identify at least three potential vulnerabilities.
- Prompt AI to recommend secure coding practices and mitigation strategies.
- Implement the suggested improvements and test the secured version.
- Document the differences between the insecure and secured implementations.

Expected Output:

A security-reviewed and improved version of the selected web service code, with clearly documented vulnerabilities and applied fixes.

Prompt:

Generate a python code a smart city system that collects traffic data from various sensors across the city. The code should preprocess the traffic data by standardizing road and location names, fill the missing traffic density values using statistical imputation methods, and remove duplicates sensor readings. Normalize speed and vehicle count metrics, and a breif summary comparing dataset quality before and after preprocessing. Finally, the code should output a cleaned and preprocessed pandas DataFrame ready for analysis.

Code:

```
# Task-04
# Generate a python code a smart city system that collects traffic data from various sensors across the city. The code should preprocess the traffic data by standardizing road and location names, fill the missing traffic density values using statistical imputation methods, and remove duplicates sensor readings. Normalize speed and vehicle count metrics, and a breif summary comparing dataset quality before and after preprocessing. Finally, the code should output a cleaned and preprocessed pandas DataFrame ready for analysis.

import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler

# Load dataset
traffic = pd.read_csv(r"C:\AIAC LAB\traffic_data_raw.csv")
print("Original data shape:", traffic.shape)
print(traffic.head())
print("\nOriginal data summary:\n", traffic.describe())

# Standardize text fields
traffic['road_name'] = traffic['road_name'].str.strip().str.title()
traffic['location'] = traffic['location'].str.strip().str.title()

# Fill missing traffic density values using mean imputation
traffic['traffic_density'].fillna(traffic['traffic_density'].mean(), inplace=True)

# Remove duplicates
traffic.drop_duplicates(inplace=True)

# Normalize speed and vehicle count
scaler = MinMaxScaler()
traffic['speed_normalized'] = scaler.fit_transform(traffic[['speed']])
traffic['vehicle_count_normalized'] = scaler.fit_transform(traffic[['vehicle_count']])

# Display preprocessed data
print("\nCleaned data shape:", traffic.shape)
print(traffic.head())
print("\nPreprocessed data summary:\n", traffic.describe())
print("\nData types:\n", traffic.dtypes)
print("\nMissing values:\n", traffic.isnull().sum())

# Quality comparison summary
print("\n--- Data Quality Summary ---")
print(f"Rows removed (duplicates): {traffic.shape[0]}")
print("Preprocessing completed successfully!")
```

Output:

```
posts = pd.read_csv(r'C:\Users\B\Desktop\social_media_posts.csv')
Original data shape: (8, 6)
   timestamp      road_name  location_name  speed  vehicle_count  traffic_density
0  2025-01-01 08:00        NH 44      City Center    65.5           120           High
1  2025-01-01 08:15         nh 44      city center    45.2           180          Very High
2  2025-01-01 08:30  State Highway 1  Suburban Area    72.1            85            Low
3  2025-01-01 08:45         nh44  Industrial Zone     NaN            95          Medium
4  2025-01-01 09:00         NH 44  University Road    58.3           150           High

Original data summary:
   speed  vehicle_count
count  7.000000      8.000000
mean   57.828571    135.000000
std    12.827277     45.747756
min    35.800000     85.000000
25%    51.750000    106.250000
50%    62.400000    120.000000
75%    65.500000    157.500000
max    72.100000    220.000000
```

Justification:

The code provided for Task-05 effectively cleans and preprocesses social media text data for sentiment analysis. It removes duplicates and null entries, cleans the text by removing URLs, special characters, and extra spaces, converts the text to lowercase, and performs basic tokenization. Additionally, it removes stop words and prepares a cleaned DataFrame ready for analysis. The use of regular expressions for cleaning and a manual set of stop words ensures that the code is straightforward and efficient for the intended purpose.

Task 5 – Exception Handling Improvement

Analyse an AI-generated Python program that does not handle runtime errors and improve its reliability.

Instructions:

- Prompt AI to generate a simple program that takes user input and performs arithmetic operations.
- Identify possible runtime errors (e.g., invalid input, division by zero).
- Modify the program to include proper try-except blocks.
- Provide meaningful error messages for invalid inputs.
- Test the improved version with valid and invalid test cases.

Expected Output:

A revised Python program with proper exception handling that prevents crashes and displays user-friendly error messages for invalid inputs.

Prompt:

Generate a python code to clean and preprocess social media text data for sentiment analysis. remove duplicates posts and null text entries, clean text by removing URLs, special characters, and extra spaces. convert text to lowercase, andperform basic tokenization. remove stop words and perform stemming or lemmatization. Finally, the code should output a cleaned and preprocessed pandas DataFrame ready for sentiment analysis.

Code:

```
# Task-05
# Generate a python code to clean and preprocess Social media text data for
import pandas as pd
import re

# Load dataset
posts = pd.read_csv(r"C:\AIAC LAB\social_media_posts.csv")
print("Original data shape:", posts.shape)
print(posts.head())

# Remove duplicates and null entries
posts.drop_duplicates(inplace=True)
posts = posts.dropna(subset=['text'])

# Define stop words manually (common English stop words)
stop_words = {
    'a', 'an', 'and', 'are', 'as', 'at', 'be', 'by', 'for', 'from',
    'has', 'he', 'in', 'is', 'it', 'its', 'of', 'on', 'or', 'that',
    'the', 'to', 'was', 'will', 'with', 'i', 'me', 'my', 'you', 'your'
}

# Function to clean text
def clean_text(text):
    # Remove URLs
    text = re.sub(r'http\S+|www\S+|https\S+', '', text, flags=re.MULTILINE)
    # Remove special characters and extra spaces
    text = re.sub(r'[^\w-zA-Z\s]', '', text)
    text = re.sub(r'\s+', ' ', text).strip()
    # Convert to lowercase
    text = text.lower()
    # Tokenize and remove stop words
    tokens = text.split()
    tokens = [word for word in tokens if word not in stop_words]
    return ' '.join(tokens)

# Apply cleaning
posts['text_cleaned'] = posts['text'].apply(clean_text)

# Display cleaned data
print("\nCleaned data shape:", posts.shape)
print(posts.head())
print("\nData types:\n", posts.dtypes)
print("\nMissing values:\n", posts.isnull().sum())
```

Output:

```
Original data shape: (10, 2)
  post_id      text
0        1  Love this new update! #GreatApp https://exampl...
1        2   This app is terrible 🤬 Never working properly!
2        3  Love this new update! #GreatApp https://exampl...
3        4                                     NaN
0        1  Love this new update! #GreatApp https://exampl...
1        2   This app is terrible 🤬 Never working properly!
2        3  Love this new update! #GreatApp https://exampl...
3        4                                     NaN
1        2   This app is terrible 🤬 Never working properly!
2        3  Love this new update! #GreatApp https://exampl...
3        4                                     NaN
4        5  Check out my new profile pic! Looking good 🔥

Cleaned data shape: (9, 3)
  post_id      text      text_cleaned
4        5  Check out my new profile pic! Looking good 🔥  check out new profile pic looking good

Cleaned data shape: (9, 3)
  post_id      text      text_cleaned
0        1  Love this new update! #GreatApp https://exampl...  love this new update greatapp
1        2   This app is terrible 🤬 Never working properly!  this app terrible never working properly
2        3  Love this new update! #GreatApp https://exampl...  love this new update greatapp
0        1  Love this new update! #GreatApp https://exampl...  love this new update greatapp
1        2   This app is terrible 🤬 Never working properly!  this app terrible never working properly
2        3  Love this new update! #GreatApp https://exampl...  love this new update greatapp
4        5  Check out my new profile pic! Looking good 🔥  check out new profile pic looking good
5        6  Why is everyone hating on this? It's fine...  why everyone hating this fine
```



```
Data types:
  post_id      int64
  text         str
  text_cleaned str
dtype: object

Missing values:
  post_id      0
  text         0
  text_cleaned 0
dtype: int64
```

Justification:

The code provided performs comprehensive cleaning and preprocessing of social media text data for sentiment analysis. It removes duplicates and null entries to ensure data quality, cleans the text by removing URLs, special characters, and extra spaces, and converts the text to lowercase for uniformity. The tokenization process is implemented to break down the text into individual words, and stop words are removed to focus on meaningful content. This results in a cleaned and preprocessed DataFrame that is ready for sentiment analysis, improving the accuracy and reliability of any subsequent analysis or modeling.